

Error Taxonomy for Pakistani English G2P

Supplementary material for *Enabling Text-to-Speech for Pakistani English: A Grapheme-to-Phoneme Pipeline and Text Corpus*.

1. What this document is

The paper reports phoneme error rate per feature class and over complete transcriptions. This document is the qualitative counterpart: it names each kind of error the four evaluated systems make, and gives one worked example per kind, drawn from the 55-sentence evaluation set.

Four systems are compared against a single reference transcription produced and verified by linguists:

Label	System
en-us	espeak-ng, en-us voice
en-gb	espeak-ng, en-gb voice
Pipe+us	the proposed pipeline, espeak en-us as fallback G2P
Pipe+gb	the proposed pipeline, espeak en-gb as fallback G2P

Every example below is given as five transcriptions in that order, reference first. A checkmark marks the systems that match the reference on the segment under discussion.

2. Top-level classification

Every error is first an edit operation against the reference:

	Deletion	Insertion	Substitution
out-of-inventory result	—	produces a phone outside the 74-phone PakE inventory	produces a phone outside the inventory
in-inventory result	reference phone absent from output	produces an inventory phone with no reference counterpart	produces a different inventory phone

Substitutions dominate in all four systems. The distinction between substitutions that land inside the inventory and substitutions that land outside it is what the out-of-inventory rate in the paper

measures: an in-inventory substitution is a wrong Pakistani English phone, an out-of-inventory substitution is a phone Pakistani English does not have at all.

Sections 3 and 4 group these operations by their phonological cause, which is the level at which they are actionable.

3. Error classes resolved by the pipeline

These are systematic failures of both espeak voices. In each case the mismatch is not an accident of one word: the segment is simply absent from the voice's phoneme set, or the voice has no rule that would produce it.

3.1 Retroflexion not applied

English alveolar /t d/ are realised as retroflex in Pakistani English. Neither espeak voice has retroflex stops.

word	reference	en-us	en-gb	Pipe+us	Pipe+gb
test	tɛst	tɛst	tɛst	tɛst ✓	tɛst ✓
road	ˌɹoːd	ˌɹoʊd	ˌɹoəd	ˌɹoːd ✓	ˌɹoːd ✓

Operation class: substitution to out-of-inventory (t, d are not PakE phones).

3.2 Dentalisation not applied

English dental fricatives are realised as dental stops.

word	reference	en-us	en-gb	Pipe+us	Pipe+gb
three	tʰɹiː	θɹiː	θɹiː	tʰɹiː ✓	tʰɹiː ✓
these	ðiːz	ðiːz	ðiːz	ðiːz ✓	ðiːz ✓

Operation class: substitution to out-of-inventory (θ, ð are not PakE phones).

3.3 Aspiration and breathy voice not represented

Six aspirated or breathy-voiced obstruents occur in Pakistani English vocabulary. espeak has none of them and produces the plain stop, or fails on the word entirely.

word	reference	en-us	en-gb	Pipe+us	Pipe+gb
Phalia	pʰɑːlɪjaː	fɛɹliə	fɛɹliəɹ	pʰɑːlɪjaː ✓	pʰɑːlɪjaː ✓
Bhasha	bʰɑːʃɑː	bæʃə	bəʃə	bʰɑːʃɑː ✓	bʰɑːʃɑː ✓

word	reference	en-us	en-gb	Pipe+us	Pipe+gb
kheer	k ^h i:ɹ	kɪɹ	kiə	k ^h i:ɹ ✓	k ^h i:ɹ ✓
Ghotki	g ^h o:ɹki:	ga:tki	gɒtki	g ^h o:ɹki: ✓	g ^h o:ɹki: ✓
chhaliya	tʃ ^h a:li:a:	si:etʃhæliyə	si:etʃhaliyə	tʃ ^h a:li:a: ✓	tʃ ^h a:li:a: ✓
jheel	dʒ ^h i:l	dʒi:l	dʒi:l	dʒ ^h i:l ✓	dʒ ^h i:l ✓

Operation class: substitution to in-inventory when the plain counterpart is produced (p for p^h); deletion of the aspiration when the voice reads the digraph as a cluster.

3.4 Velar fricatives not represented

word	reference	en-us	en-gb	Pipe+us	Pipe+gb
Khushab	xoʃa:b	kʌʃæb	kʌʃab	xoʃa:b ✓	xoʃa:b ✓
Ghazi	ɣa:zi:	ga:zi	ga:zi	ɣa:zi: ✓	ɣa:zi: ✓

Operation class: substitution to in-inventory (k and g are PakE phones; they are simply the wrong ones).

3.5 Retroflex flap not represented

word	reference	en-us	en-gb	Pipe+us	Pipe+gb
Mohenjodaro	mo:hendʒo:ɖa: ɹo:	moʊhɛndʒoʊd æɹoʊ	məʊhɛndʒəʊda .ɹəʊ	mo:hendʒo:ɖa: ɹo: ✓	mo:hendʒo:ɖa: ɹo: ✓

3.6 /w/ not realised as a labiodental

Pakistani English merges English /w/ into /v/.

word	reference	en-us	en-gb	Pipe+us	Pipe+gb
wetlands	vɛɫləndz	wɛtləndz	wɛtləndz	vɛɫləndz ✓	vɛɫləndz ✓

Operation class: substitution to out-of-inventory (w is not a PakE phone).

3.7 FACE realised as a diphthong

word	reference	en-us	en-gb	Pipe+us	Pipe+gb
state	stɛ:t	stɛɪt	stɛɪt	stɛ:t ✓	stɛ:t ✓

word	reference	en-us	en-gb	Pipe+us	Pipe+gb
failed	fe:ld	feɪld	feɪld	fe:ld ✓	fe:ld ✓

3.8 GOAT realised as a diphthong

word	reference	en-us	en-gb	Pipe+us	Pipe+gb
homes	ho:mz	hoomz	həʊmz	ho:mz ✓	ho:mz ✓

Note the two voices differ in the diphthong they produce (ou against əʊ) and the pipeline maps both to o:.

3.9 Nasalised vowels not represented

word	reference	en-us	en-gb	Pipe+us	Pipe+gb
Gujranwala	ɡʊdʒrã.vɑ:lɑ:	ɡʊdʒɹɛnwɑ:lə	ɡʊdʒɹɛnwɑ:lə	ɡʊdʒrã.vɑ:lɑ: ✓	ɡʊdʒrã.vɑ:lɑ: ✓
Chishtian	tʃɪʃtʃjã:	kɪʃʃən	kɪʃʃən	tʃɪʃtʃjã: ✓	tʃɪʃtʃjã: ✓

Both voices realise the nasalised vowel as a vowel plus a nasal consonant, or lose the syllable entirely.

3.10 Word-final short vowels not lengthened

Pakistani English lengthens a short vowel in word-final position. Neither espeak voice does.

word	reference	en-us	en-gb	Pipe+us	Pipe+gb
Karachi	kəɹɑ:tʃi:	kəɹɑ:ɹtʃi	kəɹɑ:tʃi	kəɹɑ:tʃi: ✓	kəɹɑ:tʃi: ✓
daily	dɛ:li:	deɪli	deɪli	dɛ:li: ✓	dɛ:li: ✓
country	kʌntɹi:	kʌntɹi	kʌntɹi	kʌntɹi: ✓	kʌntɹi: ✓

Karachi is resolved by the lexicon; *daily* and *country* reach the pipeline through the fallback and are resolved by the lengthening rule.

3.11 Whole-word failure on Pakistani proper nouns and loanwords

Distinct from the segment-level classes above: here the voice does not produce a wrong Pakistani phone, it produces a different word. The grapheme sequence is read by English letter-to-sound rules and the result is unrecognisable.

word	reference	en-us	en-gb	Pipe+us	Pipe+gb
Dhol	dʰo:l	di:ha:l	di:həl	dʰo:l ✓	dʰo:l ✓
Jhang	dʒʰəŋg	dʒeɪhæŋ	dʒeɪhəŋ	dʒʰəŋg ✓	dʒʰəŋg ✓
Khattak	xətək	ka:ræk	ka:tək	xətək ✓	xətək ✓
Thatta	tʰətʰtʰa:	θærə	θatə	tʰətʰtʰa: ✓	tʰətʰtʰa: ✓
Ghulam	ɣola:m	ɡjələm	ɡjələm	ɣola:m ✓	ɣola:m ✓
jirga	dʒɪrga:	dʒɜ:gə	dʒɜ:gə	dʒɪrga: ✓	dʒɪrga: ✓
dawat	ɖa:vət̪	dæwa:t	dawɒt	ɖa:vət̪ ✓	ɖa:vət̪ ✓

This class is what the evaluation lexicon exists to handle. It accounts for a disproportionate share of the baselines' insertions and deletions, because a mis-read word changes the phone count as well as the phone identities.

3.12 Text-analysis errors: abbreviations and list markers

These arise before G2P, in the normalisation of the input string.

item	reference	en-us	en-gb	Pipe+us	Pipe+gb
Rs.	.ɹu:pi:z	ɑ:ɹɪs	ɑ:ɹɪs	.ɹu:pi:z ✓	.ɹu:pi:z ✓
Mrs.	mɪsɪs	mɪsɪz	mɪsɪz	mɪsɪs ✓	mɪsɪs ✓
(ii)	tʉ:	.ɹoʊmən tu:	.ɹəʊmən tu:	tʉ: ✓	tʉ: ✓

For the Roman numeral, both voices insert a spurious word (*Roman*) in addition to mis-reading the numeral; this is the single largest source of insertions in the baselines.

4. Error classes that remain after the pipeline

These are the residual 10.5% (Pipe+us) and 10.4% (Pipe+gb). They fall into two groups: features of the fallback accent that survive the collapses, and features of Pakistani English the pipeline does not model.

4.1 Fallback-accent features that survive

Rhoticity. Pakistani English is rhotic. en-gb is not, so Pipe+gb deletes post-vocalic ɹ, en-us marks it with ə, which Pipe+us keeps.

word	reference	en-us	en-gb	Pipe+us	Pipe+gb
summer	sʌməɹ	sʌməə	sʌmə	sʌməə	sʌmə

Pipe+gb deletes the ɹ; Pipe+us substitutes ə for the sequence əɹ. Neither matches. This is the largest single residual class in both pipeline systems.

Flapping (Pipe+us only). en-us flaps intervocalic /t/, and the flap is not caught by the t → ɾ collapse.

word	reference	en-us	en-gb	Pipe+us	Pipe+gb
capital	kæpɪtəl	kæpɪrəl	kəpɪtəl	kæpɪrəl	kəpɪtəl ✓
data	də:tɑ:	deɪrə	deɪtəɪ	də:rɑ:	də:tɑ: ✓

Note that ɾ is a Pakistani English phone, so this is an in-inventory substitution: the output is a possible PakE phone in an impossible position.

TRAP vowel (Pipe+gb only). en-gb writes the TRAP vowel as a; the reference uses æ.

word	reference	en-us	en-gb	Pipe+us	Pipe+gb
traffic	tɹæfɪk	tɹæfɪk	tɹafɪk	tɹæfɪk ✓	tɹafɪk
and	ænd	ænd	and	ænd ✓	and

Because *and* recurs in almost every sentence, this class alone accounts for a large fraction of Pipe+gb's residual error.

4.2 Pakistani English features the pipeline does not model

Resistance to function-word reduction. The reference gives function words their full spelling vowels. Both espeak voices reduce them, and the pipeline has no rule that restores them.

word	reference	en-us	en-gb	Pipe+us	Pipe+gb
of	ɔ:f	ʌv	ɒv	ʌv	ɒv
from	fɹɔ:m	fɹʌm	fɹɒm	fɹʌm	fɹɒm
across	əkɹɔ:s	əkɹʌ:s	əkɹɒs	əkɹʌ:s	əkɹɒs

of is worth singling out: every occurrence in the set is a two-phone error, since both the vowel and the final consonant differ.

THOUGHT and LOT. Where the reference has ɔ:, en-us gives ɑ: and en-gb gives ɒ.

word	reference	en-us	en-gb	Pipe+us	Pipe+gb
policy	pɔ:ləsi:	pɑ:lɪsi	pɒlɪsi	pɑ:lɪsi	pɒlɪsi
historic	hɪstɔ:ɪk	hɪstɑ:ɪk	hɪstɒɪk	hɪstɔ:ɪk ✓	hɪstɒɪk

Word-boundary fusion. espeak fuses and reduces some function-word sequences into a single token. The pipeline separates correctly but inherits the fusion elsewhere.

item	reference	en-us	en-gb	Pipe+us	Pipe+gb
had been	hæd bi:n	hædbɪn	hædbɪn	hædbɪn	hædbɪn
has been	hæz bi:n	hæzbi:n	hæzbi:n	hæzbi:n	hæzbi:n

5. Summary

The classes in Section 3 are those the pipeline is designed to fix, and it fixes them without exception at the segment level. They are also the classes that make the baselines unusable for Pakistani English synthesis: a system with no retroflex stops, no dental stops, no aspirates and no velar fricatives cannot pronounce Pakistani place names, honorifics or food terms at all.

The classes in Section 4 are narrower. Most are properties of the fallback accent that a Pakistani English fallback would not have. The rest concern vowel quality in function words and unstressed syllables, which the collapses do not address. They set the agenda for the next version of the pipeline.